

California State University, East Bay

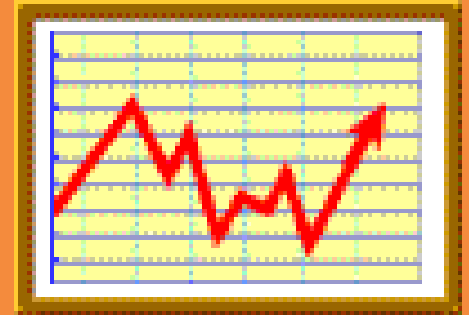
College of Business and Economics

BAN 673 Time Series Analytics

Smoothing Methods: Moving Average



Lecture Materials



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Lecture Objectives

- Define smoothing methods and their benefits
- Identify moving average approaches
- Explain how to apply centered moving average for visualization
- Utilize trailing moving average in time series forecasting
- Apply weighted moving average in time series forecasting
- Use R to develop smoothing methods for time series forecasting and for measuring forecast accuracy



Define Smoothing Methods

- **Smoothing methods** is a family of time series forecasting methods
 - **Data-driven methods** – estimate time series components (level, trend, and seasonality) directly from the data without a predetermined structure
 - **Smooth out** the noise in a series in order to uncover the data patterns
 - **Average series values** over multiple time periods with various types of smoothers, e.g., averaging different time periods or applying smoothing constants
- **Main smoothing methods**
 - **Moving average (MA)**
 - » *Centered MA, Trailing MA, Weighted MA*
 - **Simple exponential smoothing (SES)**
 - **Advanced exponential smoothing (AES)**
 - » **Holt's model** for data with trend
 - » **Winter's (Holt-Winter's) model** for data with trend and seasonality
- **Smoothing methods are useful for**
 - *Data visualization* (centered moving average)
 - *Short-term forecasting* (with trailing MA, weighted MA, and SES) for the next time period (next day, month, etc.)
 - *Two-level forecasting*, where trailing MA or SES are being used for forecasting residuals
 - *Multi-period forecasting* with advanced smoothing methods (e.g., Holt-Winter's model)

Recap: Components of Time Series and Modeling Principles

■ Time series components

- **Level** (always present)
- **Trend** – steady increase/decrease over time
- **Seasonality** – pattern that repeats itself every season
- **Random noise** (always present)

■ Time series analysis

- **Reasonable** – selected model(s) should adequately represent historical time series
- **Parsimony** – in general, the simpler the model the better
- **Goodness of fit** – find a model that fits well to the historical time series; the lower the errors (residuals) the better the model

■ Time series forecasting

- **Forecast accuracy**
- **Reasonable** and **parsimony**

Moving Average

- ***Moving average (MA)*** – common method of time series forecasting
 - *Simple smoother*
 - Forecast future points by using an ***average of several past consecutive points with window k (number of points or width of k) defined by user***
 - The term "*moving*" represents the fact that as each new actual data point becomes available, a revised MA is computed for the next data period requiring forecasting
- ***Two main types of moving averages***
 - ***Centered moving average***
 - » Based on a window centered around time t , for which moving average is identified
 - » Useful to *visualize trends*, because the averaging operation can suppress seasonality and noise, making the trend more visible
 - ***Trailing moving average***
 - » Simple and popular
 - » With different weights – ***weighted moving average (weighted MA)***

Computing a Centered MA

- Compute average of values in window (of width k), which is **centered at period t**

$$MA_t = \frac{y_{t-(k-1)/2} + \cdots + y_{t-1} + y_t + y_{t+1} + \cdots + y_{t+(k-1)/2}}{k}$$

- **Odd width:** center window on time t and average the values in the window
- **Even width:** take the two “almost centered” windows and average the values in them

Diagram illustrating the computation of a centered moving average for even and odd window widths.

Even width ($k=4$): Two overlapping windows of width 4 are shown. The first window (red) covers periods $t-2$ to $t+1$. The second window (yellow) covers periods $t-1$ to $t+2$. The formula for MA_t is the average of the averages of these two windows:

$$MA_t = \left(\frac{\frac{y_{t-2} + y_{t-1} + y_t + y_{t+1}}{4} + \frac{y_{t-1} + y_t + y_{t+1} + y_{t+2}}{4} \right) / 2$$

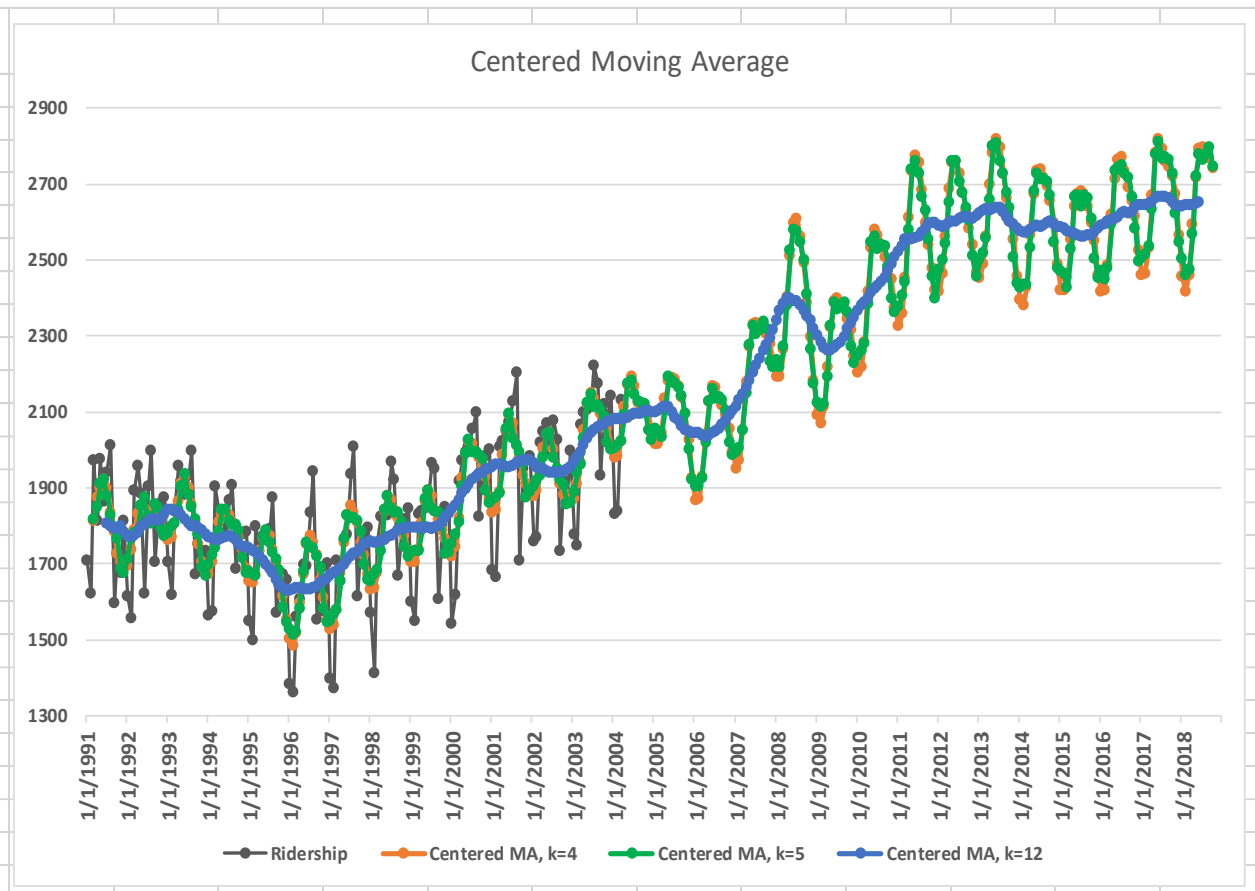
Odd width ($k=5$): A single window of width 5 is shown, covering periods $t-2$ to $t+2$. The formula for MA_t is the average of the values in this window:

$$MA_t = \frac{y_{t-2} + y_{t-1} + y_t + y_{t+1} + y_{t+2}}{5}$$

- In seasonal series, the window width k may be equal to the seasonal cycle, e.g., for Amtrak data, $k = 12$ (monthly seasonality)

Centered MA in Excel

Month	Ridership	Centered MA, k=4	Centered MA, k=5	Centered MA, k=12
1/1/1991	1708.917	#N/A	#N/A	#N/A
2/1/1991	1620.586	#N/A	#N/A	#N/A
3/1/1991	1972.715	1811.727	1817.769	#N/A
4/1/1991	1811.665	1875.204	1848.457	#N/A
5/1/1991	1974.964	1901.318	1912.312	#N/A
6/1/1991	1862.356	1922.411	1920.422	#N/A
7/1/1991	1939.86	1900.198	1877.220	1805.616
8/1/1991	2013.264	1835.605	1827.212	1799.050
9/1/1991	1595.657	1785.402	1789.874	1793.009
10/1/1991	1724.924	1727.453	1764.675	1795.626
11/1/1991	1675.667	1704.924	1684.988	1797.879
12/1/1991	1813.863	1686.341	1677.274	1784.147
1/1/1992	1614.827	1692.306	1710.534	1772.653
2/1/1992	1557.088	1737.015	1766.596	1770.440
3/1/1992	1891.223	1788.516	1780.767	1774.261
4/1/1992	1955.981	1830.496	1782.410	1782.315
5/1/1992	1884.714	1840.251	1851.654	1793.608
6/1/1992	1623.042	1846.853	1872.752	1803.907
7/1/1992	1903.309	1829.342	1822.335	1810.228
8/1/1992	1996.712	1830.110	1807.392	1816.556
9/1/1992	1703.897	1848.266	1855.104	1816.845
10/1/1992	1810.000	1827.854	1849.466	1814.618
11/1/1992	1861.601	1812.825	1791.176	1816.015
12/1/1992	1875.122	1789.062	1774.103	1828.174
1/1/1993	1705.259	1762.018	1779.445	1840.218



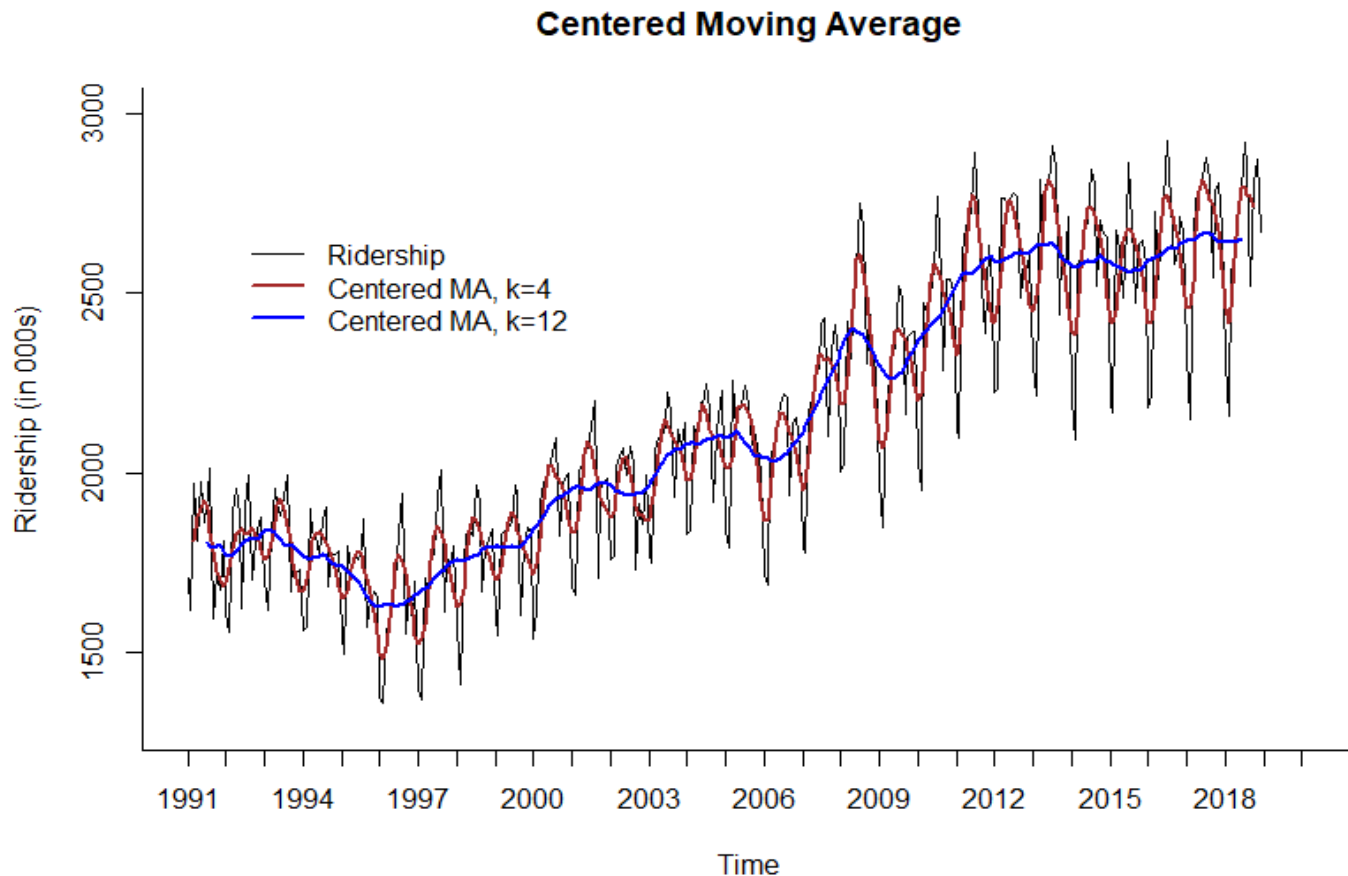
Centered MA, $k = 4$, MA at 3/1991, $t=3$:

$$MA_3 = [(1708.917+1620.586+1972.715+1811.665)/4 + (1620.586+1972.715+1811.665+1974.964)/4]/2 = 1811.727$$

Centered MA, $k=5$, MA at 3/1991, $t=3$:

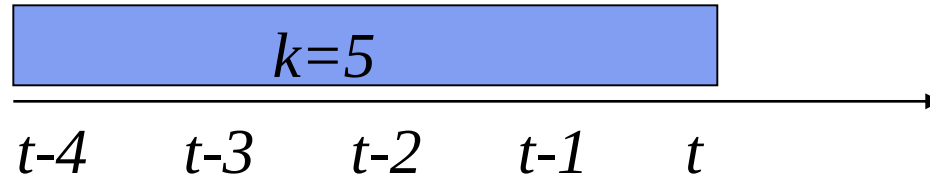
$$MA_3 = (1708.917+1620.586+1972.715+1811.665+1974.964)/5 = 1817.769$$

Visualizing Amtrak Ridership Data with Centered MA in R



Computing Trailing MA

- Choose window width (k)
- For MA at time t , place window on time points $t-k+1, t-k+2, \dots, t$

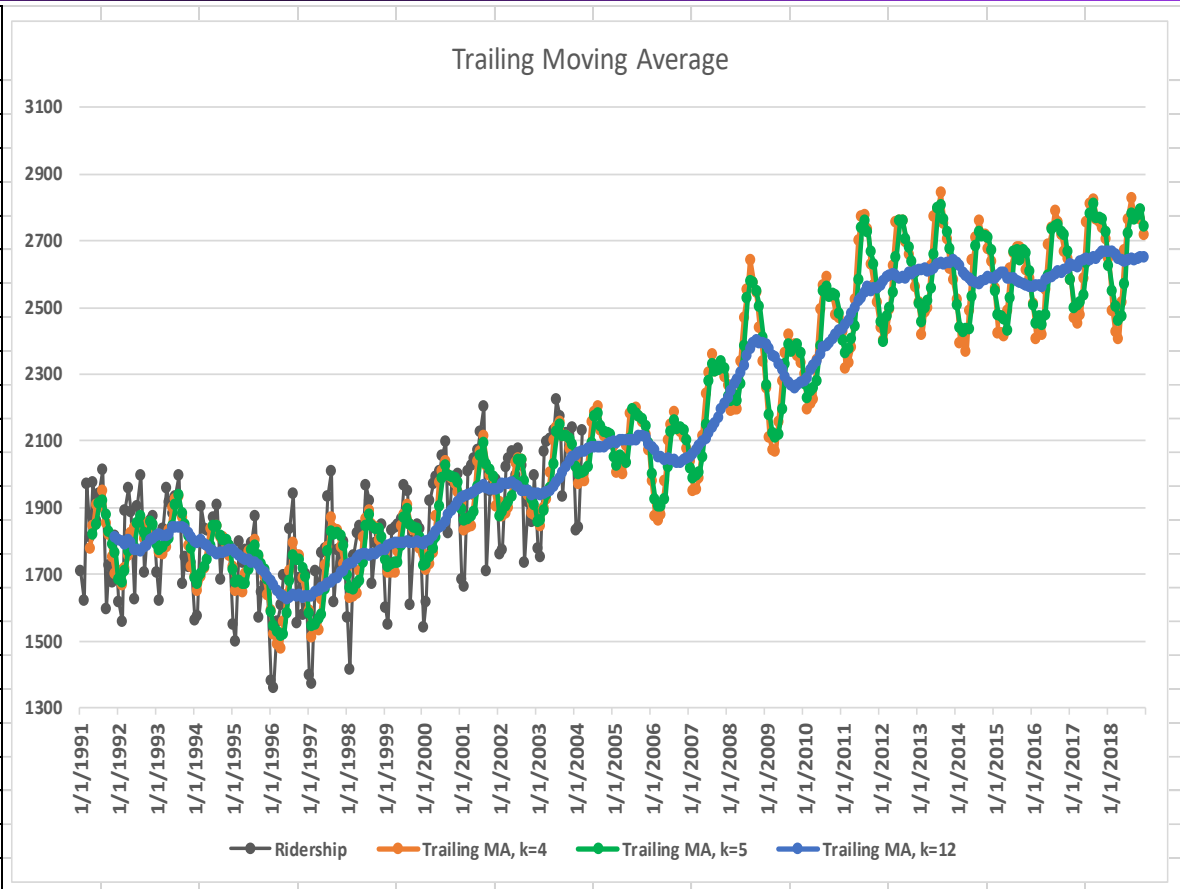


- Compute average of values in the window

$$MA_t = \frac{y_{t-k+1} + y_{t-k+2} + \dots + y_{t-1} + y_t}{k}$$

Trailing MA in Excel

Month	Ridership	Trailing MA, k=4	Trailing MA, k=5	Trailing MA, k=12	Centered MA, k=12
1/1/1991	1708.917	#N/A	#N/A	#N/A	#N/A
2/1/1991	1620.586	#N/A	#N/A	#N/A	#N/A
3/1/1991	1972.715	#N/A	#N/A	#N/A	#N/A
4/1/1991	1811.665	1778.471	#N/A	#N/A	#N/A
5/1/1991	1974.964	1844.983	1817.769	#N/A	#N/A
6/1/1991	1862.356	1905.425	1848.457	#N/A	#N/A
7/1/1991	1939.86	1897.211	1912.312	#N/A	1805.616
8/1/1991	2013.264	1947.611	1920.422	#N/A	1799.050
9/1/1991	1595.657	1852.784	1877.220	#N/A	1793.009
10/1/1991	1724.924	1818.426	1827.212	#N/A	1795.626
11/1/1991	1675.667	1752.378	1789.874	#N/A	1797.879
12/1/1991	1813.863	1702.528	1764.675	1809.537	1784.147
1/1/1992	1614.827	1707.320	1684.988	1801.696	1772.653
2/1/1992	1557.088	1665.361	1677.274	1796.404	1770.440
3/1/1992	1891.223	1719.250	1710.534	1789.613	1774.261
4/1/1992	1955.981	1754.780	1766.596	1801.640	1782.315
5/1/1992	1884.714	1822.252	1780.767	1794.119	1793.608
6/1/1992	1623.042	1838.740	1782.410	1774.176	1803.907
7/1/1992	1903.309	1841.762	1851.654	1771.130	1810.228
8/1/1992	1996.712	1851.944	1872.752	1769.751	1816.556
9/1/1992	1703.897	1806.740	1822.335	1778.771	1816.845
10/1/1992	1810.000	1853.480	1807.392	1785.860	1814.618
11/1/1992	1861.601	1843.053	1855.104	1801.355	1816.015
12/1/1992	1875.122	1812.655	1849.466	1806.460	1828.174



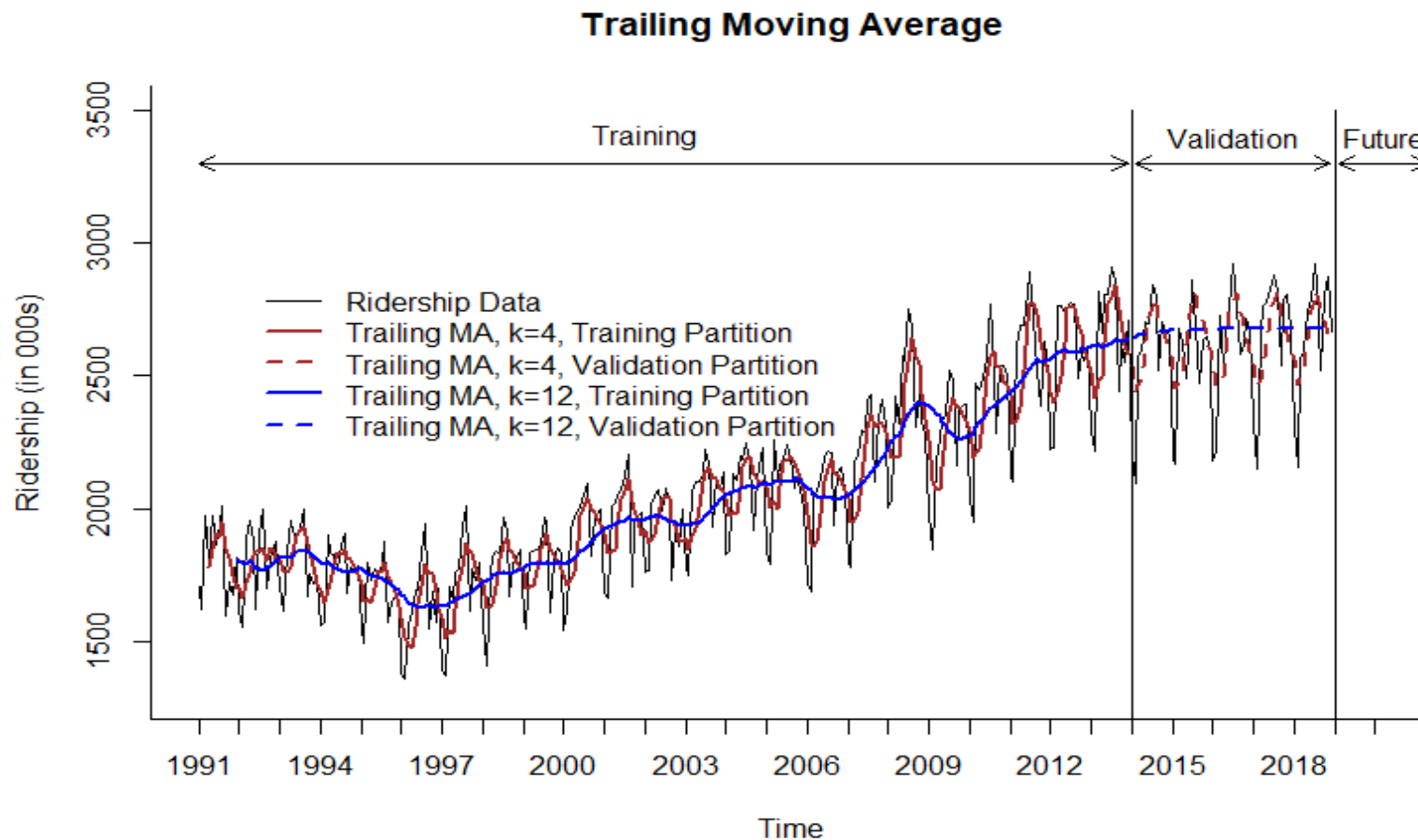
Trailing MA, k = 4, MA at 6/1991, t=6:

$$MA_6 = (1972.715 + 1811.665 + 1974.964 + 1862.356) / 4 = 1905.425$$

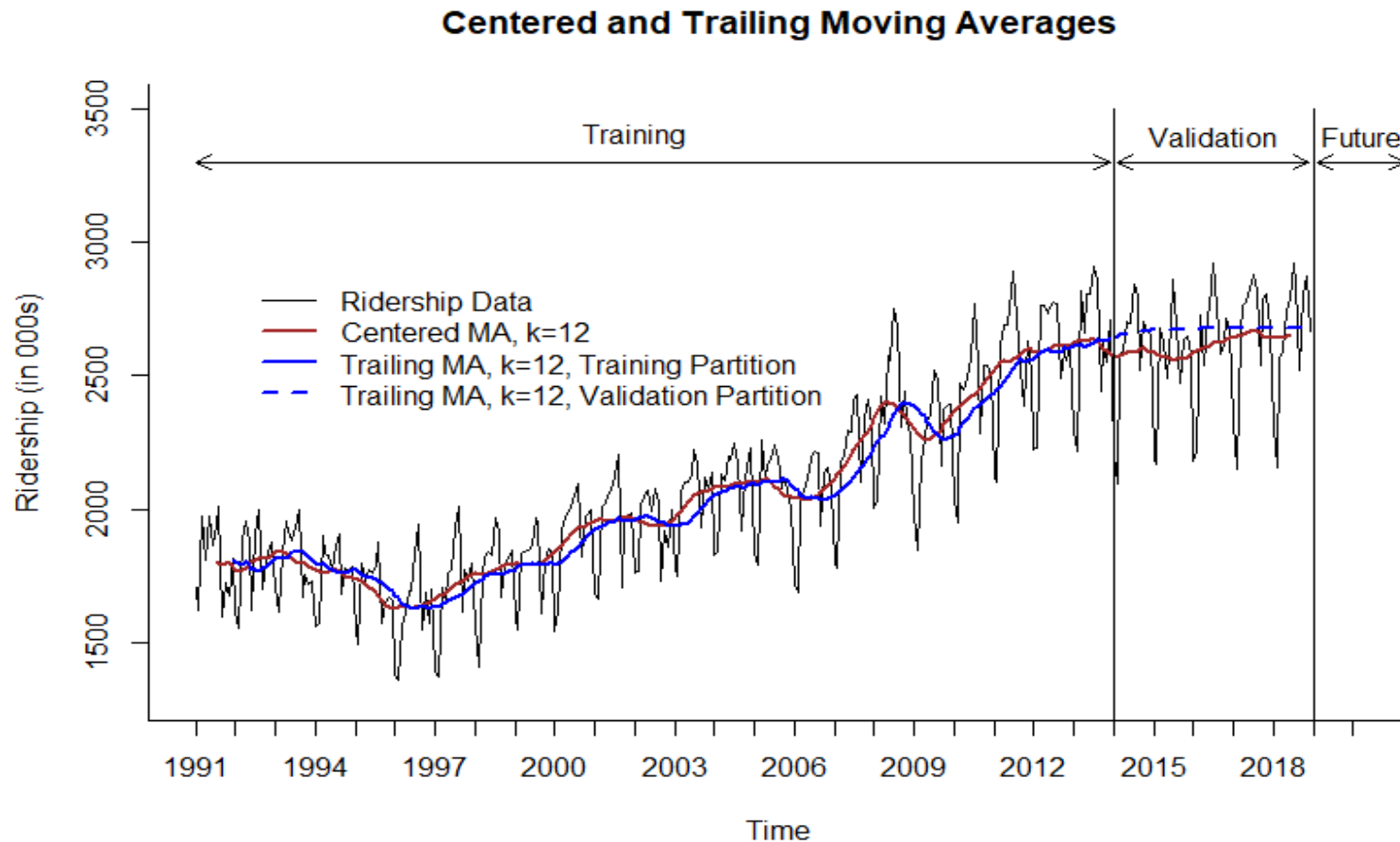
Trailing MA, k=5, MA at 6/1991, t=6:

$$MA_6 = (1620.586 + 1972.715 + 1811.665 + 1974.964 + 1862.356) / 5 = 1848.457$$

Visualizing Amtrak Ridership Data with Trailing MA for Training and Validation Periods



Visualizing Amtrak Ridership Data with Trailing MA and Trailing MA vs. Centered MA



Choosing Window Width (k)

- For trailing MA forecasting, *user must make a choice of window's width k* , which is the number of records used for averaging
- With data-driven methods, the choice of k , which is a smoothing parameter, is a *balance between under-smoothing* (narrow window, smaller k) and *over-smoothing* (wider window, larger k)
 - Wider window with larger k will make forecast smoother, and will expose more global trends
 - Narrow window with smaller k will reveal local trends
- For data with *no seasonality*, may use *narrow window* (under-smoothing)
- For data with *seasonality*, may use k *equal to the number of seasons*
- *Experiment with different values of k and compare forecast performance/accuracy based on these values*

Performance Measures for Trailing MA in Validation Partition

Trailing MA, k=4

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test Set	-17.803	168.614	136.145	-1.181	5.499	0.243	0.727

Trailing MA, k= 5

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test Set	-23.813	175.227	138.642	-1.45	5.611	0.344	0.761

Trailing MA, k=12

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test Set	-56.043	220.487	157.269	-2.897	6.48	0.452	0.968

Applying Trailing MA in Forecasting

- In general, *trailing MA should be used for forecasting in time series that lack seasonality and trend*
- In order to apply trailing MA in data with trend and/or seasonality, **two-level forecasting**, a combination of *two forecasting models*, may be applied:
 - **Level 1 - Regression model** with trend and/or seasonality. It is also used to:
 - » Remove trend (*de-trending*) and/or seasonality (*de-seasonalizing*) in historical data
 - » Identify *residuals (errors)* – differences between actual ridership and regression forecast in respective time periods
 - **Level 2 - Trailing MA** to forecast regression model's **residuals (errors)**; can be forecasted using trailing MA
- The *total forecast* used in predictions is a *combination (sum) of regression model and trailing MA forecasts*

Training Regression Model with Linear Trend and Seasonality

Call:

`tslm(formula = train.ts ~ trend + season)`

Residuals:

Min	1Q	Median	3Q	Max
-281.26	-94.77	-28.53	94.12	402.96

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1302.4480	32.2037	40.444	< 2e-16
trend	3.5132	0.1052	33.398	< 2e-16
season2	-41.0741	41.0206	-1.001	0.31760
season3	316.0566	41.0210	7.705	2.68e-13
season4	307.1209	41.0217	7.487	1.07e-12
season5	352.9515	41.0226	8.604	7.10e-16
season6	342.0432	41.0238	8.338	4.28e-15
season7	441.9060	41.0253	10.772	< 2e-16
season8	430.5533	41.0271	10.494	< 2e-16
season9	120.0197	41.0291	2.925	0.00374
season10	260.0409	41.0314	6.338	1.01e-09
season11	254.7954	41.0340	6.209	2.06e-09
season12	269.6614	41.0368	6.571	2.66e-10

trend (t) = time period index, e.g., t = 1 for Jan. 1991, t = 2 for Feb. 1992, etc.

season 2 (D_2) = binary (1,0) variable, equals to 1 if February, 0 if otherwise

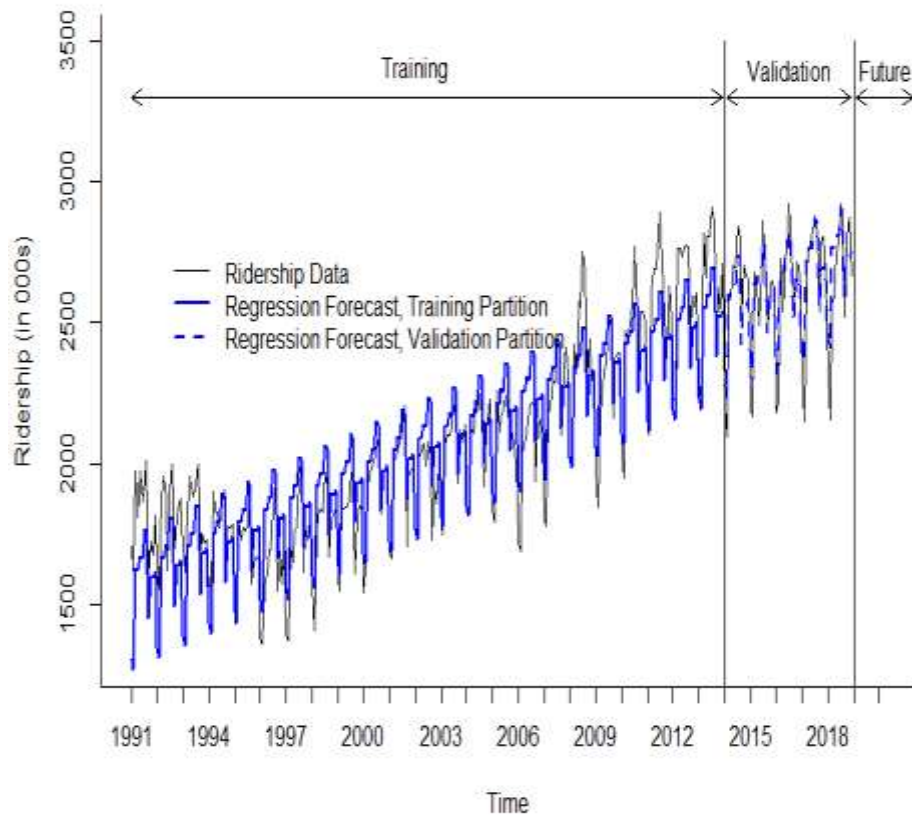
season 3 (D_3) = binary variable (1,0) equals to 1 if March, 0 if otherwise

season 12 (D_{12}) = binary variable equals to 1 if December, 0 if otherwise

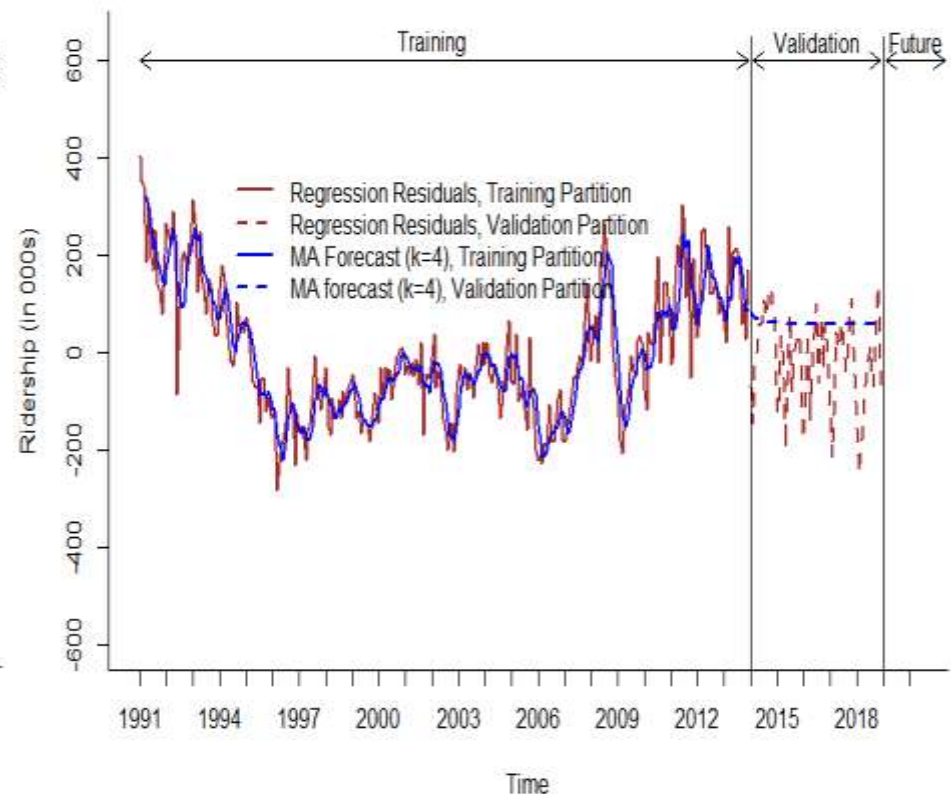
$$y_t = 1302.45 + 3.51t - 41.07 D_2 + 316.06 D_3 + \dots + 269.66 D_{12}$$

Regression and MA Trailing Forecasts in Training and Validation Partitions

Regression Forecast in Training and Validation Partitions



Regression Residuals and Trailing MA for Residuals



Two-Level Forecast for Validation Partition

- **Forecasting period from January 2014 (1) through December 2014 (12)**

	Ridership	Regression.Fst	MA.Residuals.Fst	Combined.Fst
1	2206.788	2275.614	81.94797	2357.562
2	2092.819	2238.053	77.34474	2315.398
3	2575.951	2598.697	73.66216	2672.359
4	2592.994	2593.275	70.71610	2663.991
5	2700.179	2642.618	68.35924	2710.978
6	2695.954	2635.223	66.47376	2701.697
7	2844.949	2738.599	64.96537	2803.565
8	2802.873	2730.760	63.75867	2794.519
9	2519.583	2423.740	62.79330	2486.533
10	2702.607	2567.274	62.02100	2629.295
11	2667.575	2565.542	61.40317	2626.945
12	2656.185	2583.921	60.90890	2644.830

Performance Measures for Validation Partition

*Regression
with Linear
Trend and
Seasonality*

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	-17.224	95.541	76.238	-0.892	3.04	0.42	0.424

*Regression +
Trailing MA
(k=4) for
Residuals*

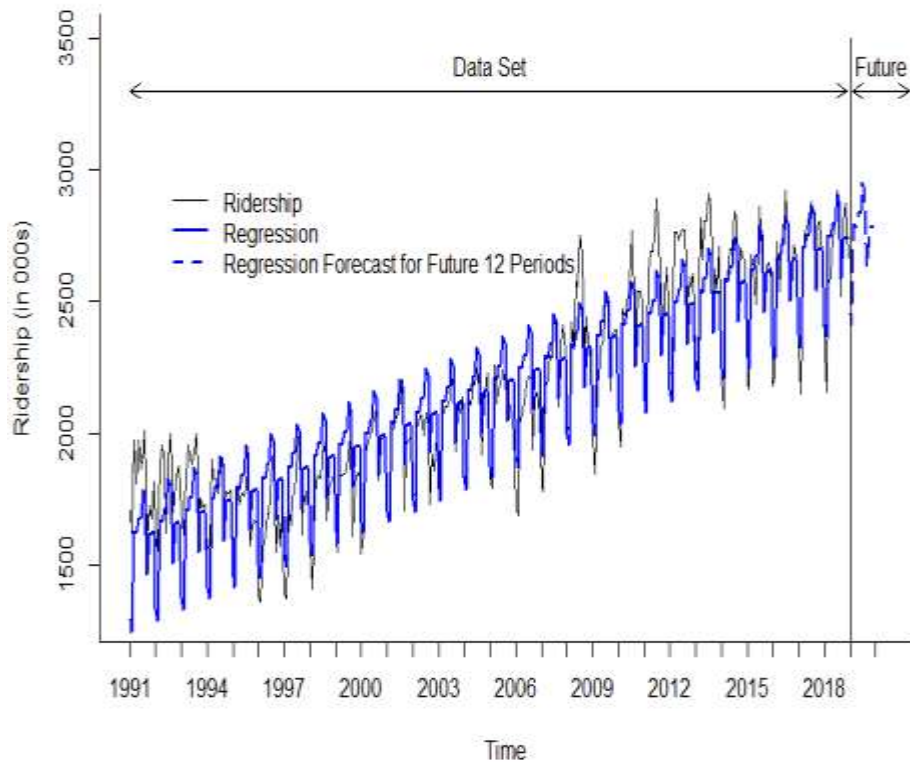
	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	-78.074	122.247	93.741	-3.239	3.805	0.418	0.543

Forecasting for Future Periods

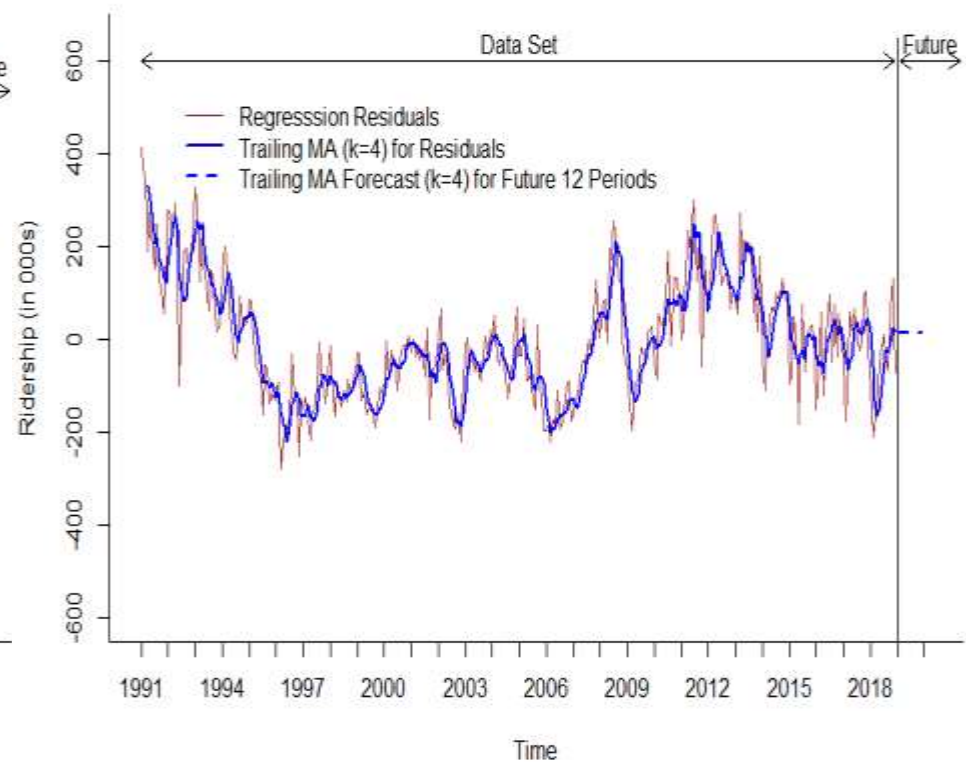
- Before attempting to forecast future values of time series, *the training and validation periods must be recombined into one (entire) time series data set*
- The chosen model or several alternative models, e.g., *regression model with linear trend and seasonality and two-level model with regression and trailing MA for residuals*, *need to be rerun on the entire (complete) data set*
- *Measure model or several models' performance vs. original data* using accuracy measures like *MAPE*, *RMSE* and others
- The best forecasting model, based on the entire data set, can be now used for forecasting future periods

Regression and Trailing MA Models for Entire Data Set and Future 12 Periods

Ridership Data and Regression with Trend and Seasonality



Regression Residuals and Trailing MA for Residuals



Two-Level Forecast for Future 12 Periods

- Forecasting period from January 2019 (1) through December 2019 (12)

	Regression.Fst	MA.Residuals.Fst	Combined.Fst
1	2456.301	15.74498	2472.046
2	2409.923	15.48028	2425.403
3	2792.287	15.26849	2807.556
4	2784.801	15.09904	2799.901
5	2836.938	14.96345	2851.902
6	2841.993	14.85497	2856.848
7	2951.144	14.76817	2965.913
8	2926.603	14.69872	2941.302
9	2628.541	14.64316	2643.184
10	2779.037	14.59870	2793.636
11	2783.325	14.56313	2797.888
12	2782.436	14.53466	2796.970

Performance Measures for Entire Data Set

*Regression
with Linear
Trend and
Seasonality*

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	0	128.744	103.733	-0.33	5.103	0.793	0.693

*Regression +
Trailing MA
(k=4) for
Residuals*

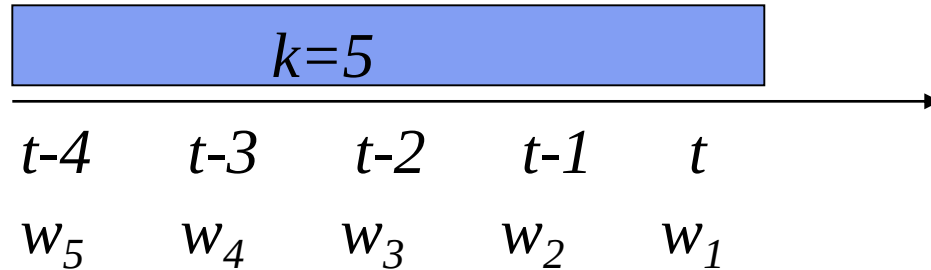
	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	-1.557	59.443	46.529	-0.15	2.211	0.259	0.302

*Seasonal
Naïve
Forecast*

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	31.113	116.076	93.101	1.26	4.375	0.631	0.579

Weighted Moving Average (WMA)

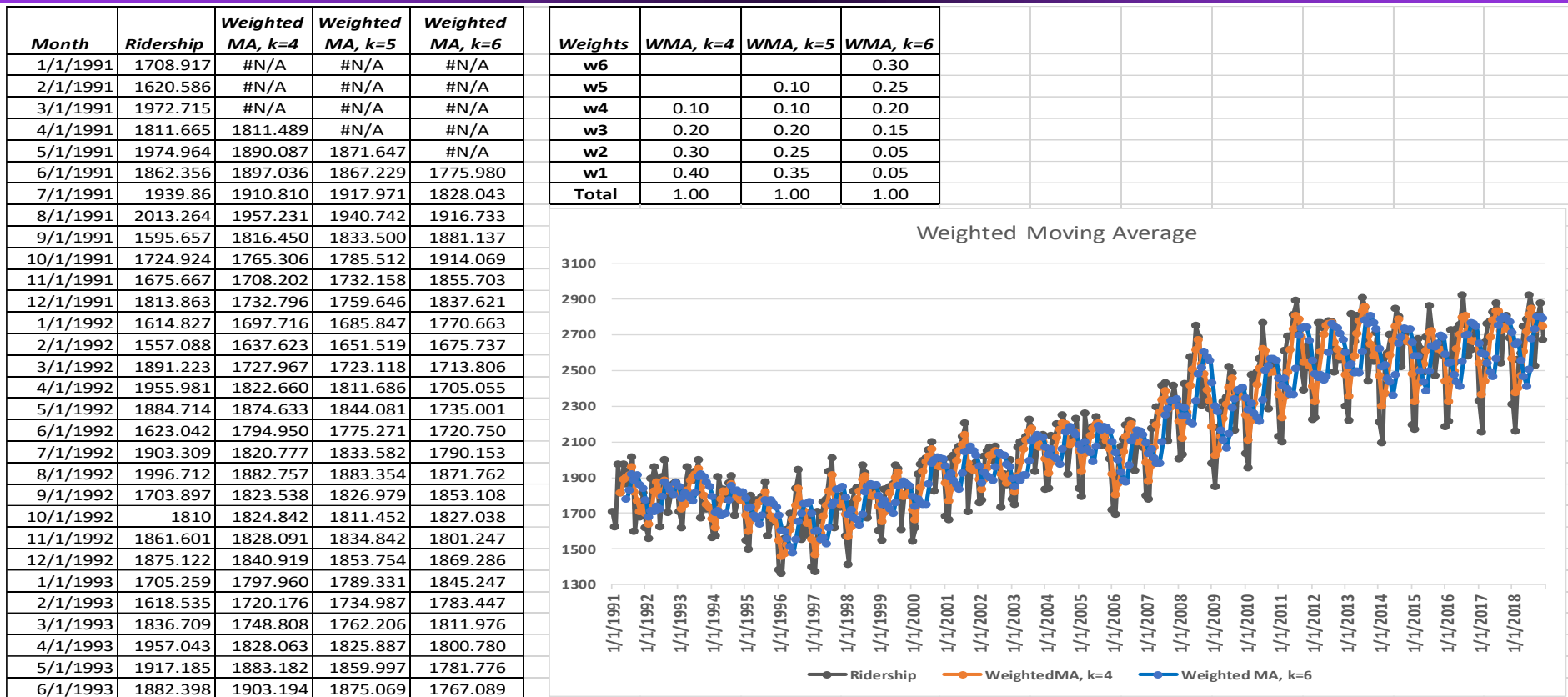
- Choose window width (k)
- For MA at time t , place window on time points $t-k+1, t-k+2, \dots, t$ and weights $w_k, w_{k-1}, \dots, w_2, w_1$ ($0 \leq w_i \leq 1, \sum_{i=1}^k w_i = 1$)



$$WMA_t = w_k y_{t-k+1} + w_{k-1} y_{t-k+2} + \dots + w_2 y_{t-1} + w_1 y_t$$

- In general, WMA can be used for forecasting only in series that lack seasonality and trend
- In order to apply moving average, it is suggestive to remove trend and/or seasonality (if they exist in the data)

Weighted MA in Excel



WMA, k = 4, WMA at 6/1991, t=6:

$$\text{WMA}_6 = 0.10 \cdot 1972.715 + 0.20 \cdot 1811.665 + 0.30 \cdot 1974.964 + 0.40 \cdot 1862.356 = 1897.036$$

WMA, k=5, WMA at 6/1991, t=6:

$$\text{WMA}_6 = 0.10 \cdot 1620.586 + 0.10 \cdot 1972.715 + 0.20 \cdot 1811.665 + 0.25 \cdot 1974.964 + 0.35 \cdot 1862.356 = 1867.229$$

Visualizing Amtrak Ridership Data with Weighted Moving Average (WMA) in R

